

A Fresh Look at Calculating the Stationary Distribution of a Markov Chain, with Applications

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Abstract

We use the random-product technique from Buckingham and Fralix (*Markov Processes and Related Fields*, 2015) to expand much further on the classical problem of determining the stationary distribution of a countable-state Markov chain, in both the discrete-time and continuous-time contexts. Namely, by focusing on the invariant measure associated with a recurrent state, we show that this invariant measure (as well as other related measures) satisfies a linear system that is slightly different from the global balance equations, yet at the same time has a structure that allows for the numerical calculation of the invariant measure (and hence, the stationary distribution) through a host of well-known numerical schemes that are typically used when Gaussian elimination becomes impractical (e.g. when the state space is very large). We suspect these results could open the door to the development of other numerical schemes for efficiently calculating stationary distributions of Markov chains.

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1 Introduction

Suppose $\{X_n\}_{n \geq 0}$ represents a discrete-time Markov chain with a countable state space S and a transition probability matrix $\mathbf{P} := [p(x, y)]_{x, y \in S}$. It is very well-known that when $\{X_n\}_{n \geq 0}$ is both irreducible and positive recurrent, it has a unique stationary distribution $\pi := [\pi(x)]_{x \in S}$, meaning π is an invariant measure of $\{X_n\}_{n \geq 0}$ that satisfies the normalizing equation

$$\sum_{x \in S} \pi(x) = 1.$$

Recall that a nonnegative row vector (i.e. a measure on S) $\gamma := [\gamma(x)]_{x \in S}$ is an invariant measure of $\{X_n\}_{n \geq 0}$ if $\gamma = \gamma \mathbf{P}$. Each equation making up $\gamma = \gamma \mathbf{P}$ is of the following form: for each $y \in S$,

$$\gamma(y) = \sum_{z \in S} \gamma(z) p(z, y) \tag{1}$$

where we refer to Equation (1) as the *global balance equation associated with state y* . By identifying each global balance equation with each state, we can define, for each nonempty subset $E \subset S$, $\mathcal{G}_E$ as the set of all global balance equations associated with the states within E : in other words, Equation (1) is an equation within $\mathcal{G}_E$ if and only if $y \in E$.

For some types of irreducible, positive recurrent DTMCs, π can be found analytically, but this is often not the case. Gaussian elimination can be used to calculate π numerically when S is small or moderately-sized, but this process becomes inefficient when S is finite but very large. In this case, iterative methods are often used, the simplest being the *power method*: pick an initial distribution $\mathbf{x}_0$ on S , then for each integer $n \geq 1$, use the recursive scheme

$$\mathbf{x}_n = \mathbf{x}_{n-1} \mathbf{P}, \quad n = 1, 2, 3, \dots$$

to calculate $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots$, and stop when the difference (depending on some predefined metric) between $\mathbf{x}_n$ and $\mathbf{x}_{n-1}$ is sufficiently small: once you stop, the estimate for π is $\mathbf{x}_n$. This method is rather naive, and convergence is only guaranteed to occur when $\{X_n\}_{n \geq 0}$ is also

aperiodic. More elaborate iterative methods have been established for DTMCs with very large state spaces, and these methods have become especially popular thanks to Google PageRank, see e.g. Langville and Meyer [12]. A recently-established iterative method is the *Red Light Green Light* method of Avrachenkov et al [3], which subsumes many known iterative methods: see also Litvak [14] where an interesting open problem associated with this method is proposed.

In this work, we take a step back, and return to the fundamental problem of finding the stationary distribution of an irreducible, positive recurrent DTMC. In the classical theory of DTMCs, the key to establishing the existence of an invariant measure for an irreducible, recurrent DTMC is to consider, for some fixed state $x \in S$, the measure $\boldsymbol{\mu}^{(x)} := [\mu^{(x)}(y)]_{y \in S}$ defined as

$$\mu^{(x)}(y) := \mathbb{E}_x \left[\sum_{n=0}^{\eta_x-1} \mathbf{1}_{\{X_n=y\}} \right], \quad y \in S$$

where $\eta_x := \inf\{n \geq 1 : X_n = x\}$, and $\mathbf{1}_{\{\cdot\}}$ denotes an indicator variable, equal to 1 if $\{\cdot\}$ is true, and 0 otherwise. While it is well-known that $\boldsymbol{\mu}^{(x)}$ is an invariant measure when x is recurrent, we will soon see that $\boldsymbol{\mu}^{(x)}$ appears as the limit of a sequence of measures $\{\boldsymbol{\gamma}_m^{(x)}\}_{m \geq 0}$ defined on S , which satisfy a simple recursive scheme. Here the initial vector $\boldsymbol{\gamma}_0^{(x)}$ satisfies $\gamma_0^{(x)}(x) = 1$ and $\gamma_0^{(x)}(y) = 0$ for each $y \neq x$. Moreover, for each integer $m \geq 0$, $\gamma_{m+1}^{(x)}(x) = 1$, and for each $y \neq x$,

$$\gamma_{m+1}^{(x)}(y) = p(x, y) + \sum_{z \neq x} \gamma_m^{(x)}(z) p(z, y).$$

These observations can be stated more elegantly with matrix notation: if we define, for each measure γ on S , each matrix $\mathbf{C} := [c(x, y)]_{x, y \in S}$, and each pair of nonempty subsets $A, B \subseteq S$,

$$\gamma[A] := [\gamma(x)]_{x \in A}, \quad \mathbf{C}[A, B] := [c(x, y)]_{x \in A, y \in B}$$

we have that for each integer $m \geq 0$,

$$\gamma_{m+1}^{(x)}[\{x\}^c] = \mathbf{P}[\{x\}, \{x\}^c] + \gamma_m^{(x)}[\{x\}^c] \mathbf{P}[\{x\}^c, \{x\}^c]$$

and moreover,

$$\boldsymbol{\mu}^{(x)}[\{x\}^c] = \mathbf{P}[\{x\}, \{x\}^c] + \boldsymbol{\mu}^{(x)}[\{x\}^c] \mathbf{P}[\{x\}^c, \{x\}^c].$$

A few interesting observations can be made. First, it is possible to show that for each $m \geq 0$, $0 \leq \gamma_m^{(x)} \leq \gamma_{m+1}^{(x)}$. Second, the $\gamma_m^{(x)}$ vectors converge pointwise to $\boldsymbol{\mu}^{(x)}$ as $m \rightarrow \infty$, regardless of whether or not $\{X_n\}_{n \geq 0}$ is aperiodic, and the convergence rate depends on the modulus of the dominant eigenvalue of the submatrix $\mathbf{P}[\{x\}^c, \{x\}^c]$ which, when $\{X_n\}_{n \geq 0}$ is irreducible and $|S| < \infty$, must be strictly less than one. In light of these observations, we suggest replacing the power method with this scheme: once a suitable $\gamma_N^{(x)}$ has been found, where N depends on some well-defined stopping criterion, an approximation for π is simply $(1/\|\gamma_N^{(x)}\|_1) \gamma_N^{(x)}$, where $\|\gamma_N^{(x)}\|_1$ is the ℓ_1 -norm of $\gamma_N^{(x)}$. Other iterative schemes can also be used to find $\boldsymbol{\mu}^{(x)}[\{x\}^c]$ as well, due to $\mathbf{P}[\{x\}^c, \{x\}^c]$ having a dominant eigenvalue whose modulus is strictly less than one when $\{X_n\}_{n \geq 0}$ is irreducible and $|S| < \infty$. Similar types of observations will be made for measures closely related to $\boldsymbol{\mu}^{(x)}$ as well, that can also play a role in the calculation of π .

This paper is organized as follows. In Section 2, we carefully study the solution set of any subset of global balance equations associated with an irreducible DTMC $\{X_n\}_{n \geq 0}$, and similar ideas are extended to the continuous-time case in Section 3. In Section 4, we explain how the results from Sections 2 and 3 provide us with many additional ways to numerically calculate stationary distributions, and we also connect these results to other results that have been established in the literature. Finally, we conclude in Section 5 by further explaining why we suspect these ideas may lead to a host of more specialized approaches for calculating stationary distributions of Markov chains.

2 Revisiting the Global Balance Equations in Discrete Time

Suppose $\{X_n\}_{n \geq 0}$ is an irreducible DTMC, having state space S and transition probability matrix $\mathbf{P}$. Our objective in this section is to carefully study the structure of the solution set of $\mathcal{G}_E$ for some nonempty, proper subset E of S . Throughout, we let $M := E^c$ (usage of M

instead of E^c will be clearer later, particularly in Section 4) and we further define the set

$$M_0 := \{z \in M : \sup_{y \in E} p(z, y) > 0\} \quad (2)$$

as the set of states in M that $\{X_n\}_{n \geq 0}$ can reach E from in one step.

It helps to rewrite each equation from $\mathcal{G}_E$ in the following manner: for each $y \in E$, the global balance equation associated with y can be expressed as

$$\gamma(y) = \sum_{z \in M_0} \gamma(z)p(z, y) + \sum_{z \in E} \gamma(z)p(z, y). \quad (3)$$

Indeed, we introduce M_0 in (2) because if $z \in M \setminus M_0$, Equation (3) shows that its associated $\gamma(z)$ term never appears in any equation within $\mathcal{G}_E$. Thus, when we speak of a solution γ of $\mathcal{G}_E$, we can think of it as being defined only on $M_0 \cup E$. We also allow for the possibility where some elements of a nonnegative solution γ of $\mathcal{G}_E$ take on the value ∞ , and throughout we follow the convention that $\infty \cdot 0$ is interpreted as being equal to zero.

We will soon find that an important set of solutions of $\mathcal{G}_E$ consists of the row vectors $\boldsymbol{\mu}_M^{(x)} := [\mu_M^{(x)}(y)]_{y \in M_0 \cup E}$, for $x \in M_0$, whose elements are defined as follows:

$$\mu_M^{(x)}(y) := \mathbb{E}_x \left[\sum_{n=0}^{\eta_M-1} \mathbf{1}_{\{X_n=y\}} \right], \quad y \in M_0 \cup E \quad (4)$$

where for each nonempty, proper subset C of S , $\eta_C := \inf\{n \geq 1 : X_n \in C\}$. We also slightly abuse notation by defining, for each state $x \in S$, $\eta_x := \eta_{\{x\}}$. Clearly each element of $\boldsymbol{\mu}_M^{(x)}$ is nonnegative, with $\mu_M^{(x)}(x) = 1$, and $\mu_M^{(x)}(y) = 0$ for each $y \in M_0 \setminus \{x\}$: in fact, $\mu_M^{(x)}(y) = 0$ for each $y \in M \setminus \{x\}$, if we were to also define $\mu_M^{(x)}(y)$ using (4) for $y \in (M_0 \cup E)^c$.

Our first objective is to show that for each $x \in M_0$, $\boldsymbol{\mu}_M^{(x)}$ is a solution of $\mathcal{G}_E$, but in order to do this we need to introduce some more notation, and introduce some new ideas. First, we define, for each $x \in M_0$, each $y \in E$, and each integer $n \geq 1$,

$$C_{n,x,y}(E) := \{(x_0, \dots, x_n) : x_0 = x, x_n = y, x_\ell \in E, 1 \leq \ell \leq n-1, \prod_{\ell=1}^n p(x_{\ell-1}, x_\ell) > 0\}$$

which represents the set of all n -step feasible paths from x to y that only visit a state within

the set M at time zero (where that state is $x \in M_0 \subseteq M$), and we let

$$C_{x,y}(E) := \bigcup_{n=1}^{\infty} C_{n,x,y}(E_k)$$

which represents the set of all finite feasible paths from x to y that only visit a state within the set M at time zero.

Next, we recall the random-product technique that was introduced in [5]. We associate with $\{X_n\}_{n \geq 0}$ another DTMC $\{\tilde{X}_n\}_{n \geq 0}$, having the same state space S , but its transition matrix $\tilde{\mathbf{P}} := [\tilde{p}(y, z)]_{y,z \in S}$ satisfies the following property:

$$\text{for each } y, z \in S, \tilde{p}(y, z) > 0 \text{ if and only if } p(z, y) > 0. \quad (5)$$

Assuming $\{X_n\}_{n \geq 0}$ is irreducible allows us to guarantee such a $\tilde{\mathbf{P}}$ matrix exists, and under this assumption, $\{\tilde{X}_n\}_{n \geq 0}$ is also irreducible. Note that (5) must also hold when $y = z$ as well.

Next, given a nonempty, proper subset C of S , we associate with $\{\tilde{X}_n\}_{n \geq 0}$ the hitting-time random variables $\tilde{\sigma}_C$, where $\tilde{\sigma}_C := \inf\{n \geq 0 : \tilde{X}_n \in C\}$, and we again follow the convention that for each state $x \in S$, $\tilde{\sigma}_x := \tilde{\sigma}_{\{x\}}$. Readers should note the slight difference between η_C and $\tilde{\sigma}_C$: the former represents the first time $\{X_n\}_{n \geq 0}$ makes a transition to a state in C , while the latter represents the first time $\{\tilde{X}_n\}_{n \geq 0}$ visits a state in C .

The following result, Proposition 2.1, gives another representation for $\mu_M^{(x)}(y)$ for each $y \in E$. This observation has been made in multiple papers featuring the random-product technique, but given its importance in what follows we again prove the result here for the reader's convenience, as this is arguably still an unknown technique to most researchers.

Proposition 2.1 *For each $x \in M_0$, and each $y \in E$, we have*

$$\mu_M^{(x)}(y) = \mathbb{E}_y \left[\mathbf{1}_{\{\tilde{\sigma}_M < \infty, \tilde{X}_{\tilde{\sigma}_M} = x\}} \prod_{\ell=1}^{\tilde{\sigma}_M} \frac{p(\tilde{X}_\ell, \tilde{X}_{\ell-1})}{\tilde{p}(\tilde{X}_{\ell-1}, \tilde{X}_\ell)} \right]. \quad (6)$$

Proof Define, for each integer $n \geq 1$,

$$\tilde{C}_{n,y,x}(E) := \{(x_0, x_1, \dots, x_n) : x_0 = y, x_n = x, x_\ell \in E, 1 \leq \ell \leq n-1, \prod_{\ell=1}^n \tilde{p}(x_{\ell-1}, x_\ell) > 0\}$$

which represents the number of feasible paths (with respect to the $\{\tilde{X}_n\}_{n \geq 0}$ process) from y to x , consisting of exactly n steps, and the first state reached outside of E is x . When we write $\mathbf{x} \in \tilde{C}_{n,y,x}(E)$, we interpret $\mathbf{x}$ as a vector of the form $\mathbf{x} := (x_0, x_1, \dots, x_n)$ where $x_0 = y$, $x_n = x$, and $x_\ell \in E$ for $1 \leq \ell \leq n-1$. Readers should note that $(x_0, x_1, \dots, x_n) \in \tilde{C}_{n,y,x}(E)$ if and only if $(x_n, x_{n-1}, \dots, x_0) \in C_{n,x,y}(E)$. Next, observe that summing over all feasible paths of $\{\tilde{X}_n\}_{n \geq 0}$ in $\tilde{C}_{y,x}(E) := \cup_{n \geq 1} \tilde{C}_{n,y,x}(E)$ yields

$$\begin{aligned}
\mathbb{E}_y \left[\mathbf{1}_{\{\tilde{\sigma}_M < \infty, \tilde{X}_{\tilde{\sigma}_M} = x\}} \prod_{\ell=1}^{\tilde{\sigma}_M} \frac{p(\tilde{X}_\ell, \tilde{X}_{\ell-1})}{\tilde{p}(\tilde{X}_{\ell-1}, \tilde{X}_\ell)} \right] &= \sum_{n=1}^{\infty} \sum_{\mathbf{x} \in \tilde{C}_{n,y,x}(E)} \left[\prod_{\ell=1}^n \frac{p(x_\ell, x_{\ell-1})}{\tilde{p}(x_{\ell-1}, x_\ell)} \right] \prod_{\ell=1}^n \tilde{p}(x_{\ell-1}, x_\ell) \\
&= \sum_{n=1}^{\infty} \sum_{\mathbf{x} \in \tilde{C}_{n,y,x}(E)} \prod_{\ell=1}^n p(x_\ell, x_{\ell-1}) \\
&= \sum_{n=1}^{\infty} \sum_{(x_n, x_{n-1}, \dots, x_0) \in C_{n,x,y}(E)} \prod_{\ell=1}^n p(x_\ell, x_{\ell-1}) \\
&= \sum_{n=1}^{\infty} \mathbb{P}_x(X_n = y, \eta_M > n) \\
&= \mathbb{E}_x \left[\sum_{n=0}^{\eta_M-1} \mathbf{1}_{\{X_n = y\}} \right]
\end{aligned}$$

which establishes (6), thus proving the claim. $\diamond$

Proposition 2.2 $\mu_M^{(x)}$ is a nonnegative solution of $\mathcal{G}_E$. Moreover, $\mu_M^{(x)}(y) < \infty$ for each $y \in M_0 \cup E$

Proof It is obvious from the definition of $\mu_M^{(x)}$ that each of its elements is nonnegative, with $\mu_M^{(x)}(x) = 1$, and $\mu_M^{(x)}(y) = 0$ for each $y \in M_0 \setminus \{x\}$. Moreover, for each $y \in E$,

$$\mu_M^{(x)}(y) = \mathbb{E}_x \left[\sum_{n=0}^{\eta_M-1} \mathbf{1}_{\{X_n = y\}} \right] \leq \mathbb{E}_x \left[\sum_{n=0}^{\eta_x-1} \mathbf{1}_{\{X_n = y\}} \right] < \infty$$

where the latter inequality is known to hold, regardless of whether or not states x and y are transient or recurrent: see e.g. [7] for details.

It remains to show $\mu_M^{(x)}$ is a solution of $\mathcal{G}_E$. From Proposition 2.1, we see that for each

$y \in E$, conditioning on $\tilde{X}_1$ yields

$$\begin{aligned}\mu_M^{(x)}(y) &= \sum_{z \in M_0 \cup E} \mathbb{E}_y \left[\mathbf{1}_{\{\tilde{\sigma}_M < \infty, \tilde{X}_{\tilde{\sigma}_M} = x\}} \prod_{\ell=1}^{\tilde{\sigma}_M} \frac{p(\tilde{X}_\ell, \tilde{X}_{\ell-1})}{\tilde{p}(\tilde{X}_{\ell-1}, \tilde{X}_\ell)} \mid \tilde{X}_1 = z \right] \tilde{p}(y, z) \\ &= p(x, y) + \sum_{z \in E} \mu_M^{(x)}(z) p(z, y)\end{aligned}$$

which proves the claim. $\diamond$

A closer look at the proof of Proposition 2.2 reveals that $\mu_M^{(x)}$ satisfies a more restrictive system of linear equations, which we denote as $\mathcal{G}_{E;x}$: we say $\gamma := [\gamma(y)]_{y \in M_0 \cup E}$ is a solution of $\mathcal{G}_{E;x}$ if it satisfies the following set of linear equations:

$$\gamma(x) = 1, \quad \gamma(z) = 0 \quad \text{for each } z \in M_0 \setminus \{x\}$$

and

$$\gamma(y) = p(x, y) + \sum_{z \in E} \gamma(z) p(z, y), \quad \text{for each } y \in E.$$

Clearly, each solution γ of $\mathcal{G}_{E;x}$ is also a solution of $\mathcal{G}_E$.

The next result shows that, for each $x \in M_0$, $\mu_M^{(x)}$ is the minimal nonnegative solution among all nonnegative solutions of $\mathcal{G}_{E;x}$. In the process of establishing this result, we will make use of a collection of row vectors $\{\gamma_m^{(x)}\}_{m \geq 0}$ on $M_0 \cup E$, which are defined in the following manner. First, $\gamma_0^{(x)}(x) = 1$, and $\gamma_0^{(x)}(y) = 0$ for each $y \in (M_0 \cup E) \setminus \{x\}$. Moreover, for each integer $m \geq 0$, $\gamma_{m+1}^{(x)}(x) = 1$, $\gamma_{m+1}^{(x)}(y) = 0$ for each $y \in M_0 \setminus \{x\}$, and for each $y \in E$,

$$\gamma_{m+1}^{(x)}(y) = p(x, y) + \sum_{z \in E} \gamma_m^{(x)}(z) p(z, y).$$

Theorem 2.1 *For each state $x \in M_0$, $\mu_M^{(x)}$ is the minimal nonnegative solution among the set of all nonnegative solutions of $\mathcal{G}_{E;x}$. Furthermore, $\lim_{m \rightarrow \infty} \gamma_m^{(x)} = \mu_M^{(x)}$ on $M_0 \cup E$.*

Proof For each $x \in M_0$, and each integer $m \geq 0$, define the row vector $\mu_{M,m}^{(x)} := [\mu_{M,m}^{(x)}(y)]_{y \in E_k}$

as follows: for each $y \in E$,

$$\mu_{M,m}^{(x)}(y) := \mathbb{E}_y \left[\mathbf{1}_{\{\tilde{\sigma}_M \leq m, \tilde{X}_{\tilde{\sigma}_M} = x\}} \prod_{\ell=1}^{\tilde{\sigma}_x} \frac{p(\tilde{X}_\ell, \tilde{X}_{\ell-1})}{\tilde{p}(\tilde{X}_{\ell-1}, \tilde{X}_\ell)} \right]$$

where the transition matrix of $\{\tilde{X}_n\}_{n \geq 0}$ satisfies the same conditions as in the proof of Proposition 2.1. Here we clearly see that $\mu_{M,m+1}^{(x)}(y) \geq \mu_{M,m}^{(x)}(y)$ for each $y \in E$, and $\lim_{m \rightarrow \infty} \mu_{M,m}^{(x)}(y) = \mu_M^{(x)}(y)$. Finally, given any nonnegative solution γ of $\mathcal{G}_{E;x}$, observe that

$$\mu_{M,1}^{(x)}(y) = p(x, y) \leq p(x, y) + \sum_{z \in E} \gamma(z) p(z, y) = \gamma(y).$$

Proceeding now by induction, assume $\mu_{M,m}^{(x)} \leq \gamma$ for some integer $m \geq 1$. Then by a conditioning argument, we get

$$\mu_{M,m+1}^{(x)}(y) = p(x, y) + \sum_{z \in E} \mu_{M,m}^{(x)}(z) p(z, y) \leq p(x, y) + \sum_{z \in E} \gamma(z) p(z, y) = \gamma(y).$$

This proves $\mu_{M,m}^{(x)} \leq \gamma$ for each integer $m \geq 1$, and taking limits reveals $\mu_M^{(x)} \leq \gamma$: in other words, $\mu_M^{(x)}$ is the minimal nonnegative solution of $\mathcal{G}_{E;x}$. The same logic also reveals that $\gamma_m^{(x)} = \mu_{M,m}^{(x)}$ for each integer $m \geq 0$, which completes the proof of the claim. $\diamond$

The proof of Theorem 2.1 is very similar to a recent proof of a well-known result from the theory of matrix-analytic methods, which says that the rate matrix of a quasi-birth-death (QBD) chain is the minimal nonnegative solution to a matrix quadratic equation. Interested readers should consult [10], where the argument is given in the continuous-time context. Indeed, one interesting question we leave for future research involves investigating to what extent the theory presented in this paper can be fruitfully applied toward the study of various types of level-dependent quasi-birth-death processes, whose stationary distributions consist of a countably infinite number of rate matrices instead of a single rate matrix: as far as we are aware, this paper marks the first time anyone has observed that the rate matrices of a level-dependent QBD chain satisfy a linear system of equations that is very similar to the global balance equations of a DTMC. Readers interested in learning

more about level-dependent quasi-birth-death processes are referred to [4, 13].

We now turn our attention toward understanding the structure of the solution set of $\mathcal{G}_E$.

Proposition 2.3 *The solutions $\{\mu_M^{(x)}\}_{x \in M_0}$ of $\mathcal{G}_E$ are linearly independent over $M_0 \cup E$.*

Proof Let $\{\alpha_x\}_{x \in M_0}$ denote any collection of real numbers satisfying

$$\sum_{x \in M_0} \alpha_x \mu_M^{(x)}(y) = 0$$

for each $y \in M_0 \cup E$. Choose an arbitrary $x \in M_0$, and observe that since $\mu_M^{(z)}(x) = 0$ for each $z \in M_0 \setminus \{x\}$, we get

$$0 = \sum_{z \in M_0} \alpha_z \mu_M^{(z)}(x) = \alpha_x \mu_M^{(x)}(x) = \alpha_x$$

which proves the claim. $\diamond$

The next result shows that, when $\{X_n\}_{n \geq 0}$ is irreducible and positive recurrent, each solution $\mu_M^{(x)}$, $x \in M_0$, is also a member of $\mathcal{V}_E$, the set of all absolutely convergent solutions of $\mathcal{G}_E$.

Proposition 2.4 *Suppose $\{X_n\}_{n \geq 0}$ is irreducible and positive recurrent. Then for each $x \in M_0$, $\mu_M^{(x)}$ has finite total mass on $M_0 \cup E$, meaning*

$$\sum_{y \in M_0 \cup E} \mu_M^{(x)}(y) < \infty.$$

In other words, $\mu_M^{(x)}$ is an element of $\mathcal{V}_E$.

Proof First, notice that

$$\sum_{y \in M_0 \cup E} \mu_M^{(x)}(y) = 1 + \sum_{y \in E} \mathbb{E}_x \left[\sum_{n=0}^{\eta_M-1} \mathbf{1}_{\{X_n=y\}} \right] = 1 + \mathbb{E}_x[\eta_M - 1] \leq 1 + \mathbb{E}_x[\eta_x] - 1 = \mathbb{E}_x[\eta_x] < \infty$$

which proves $\mu_M^{(x)}$ has finite total mass on $M_0 \cup E$, and since $\mu_M^{(x)}$ is nonnegative, $\mu_M^{(x)} \in \mathcal{V}_E$.

◇

Our next result shows that if two elements of $\mathcal{V}_E$ agree on M_0 , they must also agree on $M_0 \cup E$. This observation will help us generate a basis for $\mathcal{V}_E$ when $\{X_n\}_{n \geq 0}$ is irreducible and positive recurrent.

Proposition 2.5 *Suppose $\{X_n\}_{n \geq 0}$ is irreducible, and suppose $\mu_1, \mu_2 \in \mathcal{V}_E$. If $\mu_1(x) = \mu_2(x)$ for each $x \in M_0$, then $\mu_1 = \mu_2$.*

Proof First observe that for each $y \in E$,

$$\mu_1(y) - \mu_2(y) = \sum_{z \in M_0} (\mu_1(z) - \mu_2(z))p(z, y) + \sum_{z \in E} (\mu_1(z) - \mu_2(z))p(z, y) = \sum_{z \in E} (\mu_1(z) - \mu_2(z))p(z, y)$$

which further implies, for each integer $n \geq 1$,

$$|\mu_1(y) - \mu_2(y)| = \left| \sum_{z \in E} (\mu_1(z) - \mu_2(z)) \mathbb{P}_z(X_n = y, \eta_M > n) \right|.$$

Next, observe that if all states are recurrent, then for each $z, y \in E$ (where possibly $z = y$)

$$\limsup_{n \rightarrow \infty} \mathbb{P}_z(X_n = y, \eta_M > n) \leq \limsup_{n \rightarrow \infty} \mathbb{P}_z(\eta_M > n) = 0.$$

Likewise, if all states are transient, then

$$\limsup_{n \rightarrow \infty} \mathbb{P}_z(X_n = y, \eta_M > n) \leq \limsup_{n \rightarrow \infty} \mathbb{P}_z(X_n = y) = 0.$$

In light of these observations, applying the dominated convergence theorem (valid since $\mu_1, \mu_2 \in \mathcal{V}_E$) yields

$$|\mu_1(y) - \mu_2(y)| = \lim_{n \rightarrow \infty} \left| \sum_{z \in E} (\mu_1(z) - \mu_2(z)) \mathbb{P}_z(X_n = y, \eta_M > n) \right| = 0$$

proving $\mu_1(y) = \mu_2(y)$ for each $y \in E$, from which it follows that $\mu_1 = \mu_2$. ◇

Corollary 2.1 *Suppose $\{X_n\}_{n \geq 0}$ is both irreducible and positive recurrent. Then the collection of solutions $\{\mu_M^{(x)}\}_{x \in M_0}$ forms a basis of $\mathcal{V}_E$.*

Proof Let μ be an arbitrary element of $\mathcal{V}_E$. Proposition 2.4 guarantees the collection $\{\mu_M^{(x)}\}_{x \in M_0}$ consist of members of $\mathcal{V}_E$, and there is a unique linear combination $\bar{\mu}$ of $\{\mu_M^{(x)}\}_{x \in M_0}$ satisfying $\bar{\mu}(z) = \mu(z)$ for each $z \in M_0$. Furthermore, since $\bar{\mu} \in \mathcal{V}_E$ and $\bar{\mu} = \mu$ on M_0 , it follows from Proposition 2.5 that $\bar{\mu} = \mu$ on $M_0 \cup E$, which proves the collection $\{\mu_M^{(x)}\}_{x \in M_0}$ spans $\mathcal{V}_E$. This fact, combined with Proposition 2.3, proves this collection forms a basis of $\mathcal{V}_E$. $\diamond$

A similar set of results can be established that provide information on how likely $\{X_n\}_{n \geq 0}$ is to first visit M via a state $x \in M$, given it starts at state y . For each nonempty, proper subset C of S , define $\sigma_C := \inf\{n \geq 0 : X_n \in C\}$: we again follow the convention where for each $x \in M$, $\sigma_x := \sigma_{\{x\}}$. Next, define

$$N := \{x \in M : \sup_{z \in E} p(z, x) > 0\}$$

which represents the set of all states in M that can be reached in one step from a state in E . Using N , we also define the linear system $\mathcal{H}_E$, which represents the set of all linear equations

$$\delta(y) = \sum_{z \in N} p(y, z)\delta(z) + \sum_{z \in E} p(y, z)\delta(z), \quad y \in E$$

where a solution $\delta := [\delta(y)]_{y \in N \cup E}$ of these equations is defined on $N \cup E$. Likewise, for each $x \in N$, we define $\mathcal{H}_{E;x}$ as the set all linear equations

$$\delta(x) = 1, \quad \delta(z) = 0 \text{ for each } z \in N \setminus \{x\}$$

and

$$\delta(y) = p(y, x) + \sum_{z \in E} p(y, z)\delta(z), \quad y \in E.$$

Next, for each $x \in N$, define the vector $\mathbf{g}_M^{(x)} := [g_M^{(x)}(y)]_{y \in N \cup E}$ as follows:

$$g_M^{(x)}(y) = \mathbb{P}_y(\sigma_M < \infty, X_{\sigma_M} = x), \quad y \in N \cup E.$$

Obviously, $g_M^{(x)}(x) = 1$ and $g_M^{(x)}(z) = 0$ for each $z \in N \setminus \{x\}$. It is easy to show, via first-step analysis, that $\mathbf{g}_M^{(x)}$ is a solution of $\mathcal{H}_{E;x}$, and the vectors $\{\mathbf{g}_M^{(x)}\}_{x \in N}$ are all solutions of $\mathcal{H}_E$. In fact, it is also easy to show that $\mathbf{g}_M^{(x)}$ is the minimal nonnegative solution of $\mathcal{H}_{E;x}$, and appears as a pointwise limit of a sequence of vectors: this is summarized in the following theorem.

Theorem 2.2 *The vector $\mathbf{g}_M^{(x)}$ is the minimal nonnegative solution of $\mathcal{H}_{E;x}$. Moreover, $\lim_{m \rightarrow \infty} \delta_m^{(x)} = \mathbf{g}_M^{(x)}$, where the sequence of vectors $\{\delta_m^{(x)}\}_{m \geq 0}$ satisfy the following recursive scheme: $\delta_0^{(x)}(x) = 1$, $\delta_0^{(x)}(z) = 0$ for each $z \in (N \cup E) \setminus \{x\}$, and for each $m \geq 0$,*

$$\delta_{m+1}^{(x)}(x) = 1, \quad \delta_{m+1}^{(x)}(z) = 0 \text{ for each } z \in N$$

and for each $y \in E$,

$$\delta_{m+1}^{(x)}(y) = p(y, x) + \sum_{z \in E} p(y, z) \delta_m^{(x)}(z).$$

Proof We omit the proof, as it follows from first-step analysis, and is similar to the proof of Theorem 2.1. $\diamond$

The following result is analogous to Proposition 2.5, except now we are interested in studying $\mathcal{W}_E$, the set of all uniformly bounded solutions of $\mathcal{H}_E$. Note that, unlike in Proposition 2.5, we now have to assume $\{X_n\}_{n \geq 0}$ is both irreducible and recurrent.

Proposition 2.6 *Suppose $\{X_n\}_{n \geq 0}$ is irreducible and recurrent, and let $\mathbf{g}_1, \mathbf{g}_2 \in \mathcal{W}_E$. If $g_1(x) = g_2(x)$ for each $x \in N$, then $\mathbf{g}_1 = \mathbf{g}_2$ on $N \cup E$.*

Proof Since $\mathbf{g}_1$ and $\mathbf{g}_2$ are uniformly bounded, there exists a constant $C > 0$ such that

$|g_1(z) - g_2(z)| \leq C$ for all $z \in \tilde{N} \cup E$. Thus, for any $y \in E$, we have that

$$\begin{aligned} |g_1(y) - g_2(y)| &= \left| \sum_{z \in E} \mathbb{P}_y(X_n = z, \eta_M > n) (g_1(z) - g_2(z)) \right| \\ &\leq C \sum_{z \in E} \mathbb{P}_y(X_n = z, \eta_M > n) = C \mathbb{P}_y(\eta_M > n) \end{aligned}$$

which goes to 0 as $n \rightarrow \infty$ since $\{X_n\}_{n \geq 0}$ is recurrent. Hence $g_1(y) = g_2(y)$ for all $y \in E$, proving $\mathbf{g}_1 = \mathbf{g}_2$. $\diamond$

It is at this point not hard to see that the set of solutions $\{\mathbf{g}_M^{(x)}\}_{x \in N}$ form a linearly independent set on $N \cup E$, proven in an identical manner to Proposition 2.3. Therefore, in conjunction with the previous result, we have a result that is analogous to Corollary 2.1.

Corollary 2.2 *Let $\{X_n\}_{n \geq 0}$ be irreducible and recurrent. Then, $\{\mathbf{g}_M^{(x)}\}_{x \in N}$ forms a basis for $\mathcal{W}_E$.*

3 The Continuous-Time Case

Many of the results presented in the previous section extend naturally to the continuous-time context. Suppose $\{X(t); t \geq 0\}$ is an irreducible CTMC having a countable state space S and transition rate matrix $\mathbf{Q} := [q(x, y)]_{x, y \in S}$, where the off-diagonal elements of $\mathbf{Q}$ are nonnegative, and for each state $x \in S$,

$$q(x, x) := - \sum_{y \in S \setminus \{x\}} q(x, y) < \infty$$

meaning the elements found in each row of $\mathbf{Q}$ sum to zero. Recall that for each $x \in S$, whenever $\{X(t); t \geq 0\}$ visits state x , it spends an exponentially distributed with rate $q(x) := -q(x, x)$ amount of time in that state, and at the end of each such visit the chain makes a transition to state $y \neq x$ with probability $p(x, y) := q(x, y)/q(x)$, independently of how long it just spent in state x . Given we are assuming $\{X(t); t \geq 0\}$ is irreducible, we have $q(x) > 0$ for each $x \in S$, which means $\{X(t); t \geq 0\}$ has no absorbing states.

The transition times of $\{X(t); t \geq 0\}$ are represented by the increasing sequence of random variables $\{T_n\}_{n \geq 0}$, where $T_0 := 0$ and for each integer $n \geq 1$, T_n denotes the n th transition time of $\{X(t); t \geq 0\}$. These random variables give rise to the sequence of random variables $\{X_n\}_{n \geq 0}$, referred to as the *embedded DTMC* associated with $\{X(t); t \geq 0\}$,

where $X_n := X(T_n)$ for each integer $n \geq 0$. Note that each path of $\{X(t); t \geq 0\}$ is right-continuous and piecewise constant on $[0, \infty)$, and the set $\{T_n\}_{n \geq 1}$ corresponds to the set of discontinuity points of $\{X(t); t \geq 0\}$. Furthermore, $\{X_n\}_{n \geq 0}$ is a DTMC having transition matrix $\mathbf{P} := [p(x, y)]_{x, y \in S}$, where $p(x, x) = 0$ for each $x \in S$, and $p(x, y) = q(x, y)/q(x)$ for each $x, y \in S$ satisfying $x \neq y$.

We again express S as $S := M \cup E$, where M and E are pairwise disjoint, nonempty subsets of S .

Recall that a nonnegative row vector $\nu := [\nu(y)]_{y \in S}$ is an *invariant measure* of $\{X(t); t \geq 0\}$ if ν satisfies the following linear system of equations: for each $y \in S$,

$$\nu(y)q(y) = \sum_{z \in S \setminus \{y\}} \nu(z)q(z, y) \quad (7)$$

where we refer to Equation (7) as the global balance equation corresponding to state y . We let $\mathcal{G}_E^c$ denote the set of global balance equations associated with states in E , where the ‘ c ’ featured in the superscript of $\mathcal{G}_E^c$ refers to $\{X(t); t \geq 0\}$ being a process indexed by $[0, \infty)$, i.e. ‘continuous’ time. It is important to note how we have isolated the $\nu(y)q(y)$ term from the rest of the terms in (7), as this will be important later when we wish to allow for an invariant measure ν to satisfy the property where $\nu(y) = \infty$ for some states $y \in S$.

Our main objective in this section is to carefully study the solution set of $\mathcal{G}_E^c$. Defining

$$M_0 := \{x \in M : \sup_{y \in E} q(x, y) > 0\}$$

we note that the global balance equation associated with a state $y \in E$ can be expressed as

$$\nu(y)q(y) = \sum_{z \in M_0} \nu(z)q(z, y) + \sum_{z \in E, z \neq y} \nu(z)q(z, y)$$

meaning we can interpret each solution ν of $\mathcal{G}_E^c$ as being defined on only $M_0 \cup E$, as the values ν take outside of this set do not affect whether or not ν is a solution of $\mathcal{G}_E^c$. Analogously to what was done in the previous section, while we study the solution set of $\mathcal{G}_E^c$, it will be advantageous to study, for each $x \in M_0$, the solution set of $\mathcal{G}_{E;x}^c$, which is the linear

system consisting of the equations

$$\begin{aligned}\gamma(x) &= 1 \\ \gamma(z) &= 0 \text{ for each } z \in M_0 \setminus \{x\} \\ \gamma(y)q(y) &= q(x, y) + \sum_{z \in E \setminus \{y\}} \gamma(z)q(z, y), \text{ for each } y \in E.\end{aligned}$$

In this section, when we refer to the linear systems $\mathcal{G}_E$ and $\mathcal{G}_{E;x}$, these will correspond to the linear systems defined in the previous section, but the transition probabilities making up these linear systems will correspond to those found in the embedded DTMC $\{X_n\}_{n \geq 0}$ associated with $\{X(t); t \geq 0\}$.

In order to find and characterize solutions of $\mathcal{G}_E^c$, we follow an idea analogous to what was done in the previous section by constructing, for each $x \in M_0$, the row vector $\nu_M^{(x)} := [\nu_M^{(x)}(y)]_{y \in M_0 \cup E}$ where for each $y \in M_0 \cup E$,

$$\nu_M^{(x)}(y) := q(x) \mathbb{E}_x \left[\int_0^{\tau_M} \mathbf{1}_{\{X(t)=y\}} dt \right],$$

where $\tau_M := \inf\{t > 0 : \lim_{s \uparrow t} X(s) \neq X(t) \in M\}$. Note that $\nu_M^{(x)}(x) = 1$ and $\nu_M^{(x)}(z) = 0$ for $z \in M_0 \setminus \{x\}$.

The next result shows how the vector $\nu_M^{(x)}$ associated with $\{X(t); t \geq 0\}$ relates to the vector $\mu_M^{(x)}$ associated with the embedded DTMC $\{X_n\}_{n \geq 0}$. It is analogous to Lemma 31 on page 259 of Serfozo [18], but in the interest of making the paper easier to follow, we provide a complete proof of the result, as it is simple.

Proposition 3.1 *For each $x \in M_0$, and each $y \in E$,*

$$\nu_M^{(x)}(y) = \frac{q(x)}{q(y)} \mu_M^{(x)}(y). \quad (8)$$

Proof Fix a state $x \in M_0$, and a state $y \in E$. Then

$$\begin{aligned}
q(x)\mathbb{E}_x \left[\int_0^{\tau_M} \mathbf{1}_{\{X(t)=y\}} dt \right] &= q(x)\mathbb{E}_x \left[\sum_{k=0}^{\eta_M-1} \int_{T_k}^{T_{k+1}} \mathbf{1}_{\{X(t)=y\}} dt \right] \\
&= q(x)\mathbb{E}_x \left[\sum_{k=0}^{\eta_M-1} (T_{k+1} - T_k) \mathbf{1}_{\{X_k=y\}} \right] \\
&= q(x) \sum_{k=0}^{\infty} \mathbb{E}_x[(T_{k+1} - T_k) \mathbf{1}_{\{X_k=y, \eta_M > k\}}] \\
&= \frac{q(x)}{q(y)} \sum_{k=0}^{\infty} \mathbb{P}_x(X_k = y, \eta_M > k) \\
&= \frac{q(x)}{q(y)} \mathbb{E}_x \left[\sum_{k=0}^{\eta_M-1} \mathbf{1}_{\{X_k=y\}} \right] = \frac{q(x)}{q(y)} \mu_M^{(x)}(y)
\end{aligned}$$

which establishes (8). $\diamond$

Readers may question why we are including the $q(x)$ term in our definition of $\nu_M^{(x)}$, but this will become apparent in the next section when we further explore applications of this theory.

The following proposition establishes a useful connection between, for each $x \in M_0$, solutions of $\mathcal{G}_{E;x}^c$ and solutions of $\mathcal{G}_{E;x}$.

Proposition 3.2 *The row vector γ is a solution of $\mathcal{G}_{E;x}^c$ if and only if the row vector μ , whose elements are defined as*

$$\mu(y) := \frac{q(y)\gamma(y)}{q(x)}, \quad y \in M_0 \cup E \quad (9)$$

is a solution of $\mathcal{G}_{E;x}$. Furthermore, $\nu_M^{(x)}$ is a solution of $\mathcal{G}_{E;x}^c$.

Proof Suppose γ is a solution of $\mathcal{G}_{E;x}^c$. Then for each $y \in E$,

$$\begin{aligned}
\mu(y) &:= \frac{\gamma(y)q(y)}{q(x)} = \frac{1}{q(x)} \left[q(x, y) + \sum_{z \in E \setminus \{y\}} \gamma(z)q(z, y) \right] \\
&= p(x, y) + \frac{1}{q(x)} \sum_{z \in E \setminus y} \frac{q(x)\mu(z)}{q(z)} q(z, y) \\
&= p(x, y) + \sum_{z \in E \setminus \{y\}} \mu(z)p(z, y)
\end{aligned}$$

which proves μ is a solution of $\mathcal{G}_{E;x}$, and a similar argument can be used to show that γ is a solution of $\mathcal{G}_{E;x}^c$ when μ is a solution of $\mathcal{G}_{E;x}$. From this fact, it follows that $\nu_M^{(x)}$ is a solution of $\mathcal{G}_{E;x}^c$, because $\mu_M^{(x)}$ is a solution of $\mathcal{G}_{E;x}$. $\diamond$

Our next result shows that for each $x \in M_0$, $\nu_M^{(x)}$ is the minimal nonnegative solution to $\mathcal{G}_{E;x}^c$.

Theorem 3.1 *For each state $x \in M_0$, $\nu_M^{(x)}$ is the minimal nonnegative solution among the set of all nonnegative solutions of $\mathcal{G}_{E;x}^c$.*

Proof Let γ be a solution to $\mathcal{G}_{E;x}^c$. It follows, as a consequence of Proposition 3.2, that the μ defined in (9) is a solution of $\mathcal{G}_{E;x}$, and by Theorem 2.1, we have $\mu_M^{(x)}(y) \leq \mu(y)$ for all $y \in E$, from which we conclude via Proposition 3.1 that $\nu_M^{(x)}(y) \leq \gamma(y)$ for all $y \in E$. This proves the claim. $\diamond$

Proposition 3.3 *The solutions $\{\nu_M^{(x)}\}_{x \in M_0}$ of $\mathcal{G}_E^c$ are linearly independent.*

Proof The proof of this claim is virtually identical to the proof we gave for Proposition 2.3, and is omitted. $\diamond$

We are now ready to construct a basis for $\mathcal{V}_E^c$, the set of all absolutely convergent solutions of $\mathcal{G}_E^c$. The next proposition shows that for $x \in M_0$, $\nu_M^{(x)}$ has finite mass when $\{X(t); t \geq 0\}$ is both irreducible and positive recurrent.

Proposition 3.4 *Suppose $\{X(t); t \geq 0\}$ is both irreducible and positive recurrent. Then, $\nu_M^{(x)}$ has finite total mass on E .*

Proof Summing over the elements of $\nu_M^{(x)}$ gives

$$\sum_{y \in E} \nu_M^{(x)}(y) = \sum_{y \in E} q(x) \mathbb{E}_x \left[\int_0^{\tau_M} \mathbf{1}_{\{X(t)=y\}} dt \right] = q(x) \mathbb{E}_x [\tau_M - T_1] \leq q(x) \mathbb{E}_x [\tau_x] - 1 < \infty$$

proving the claim. $\diamond$

The next result shows that absolutely convergent solutions to $\mathcal{G}_E^c$ are uniquely determined by their values on the set M_0 . Note that in order to prove this result, we need to force the sojourn rates $\{X(t); t \geq 0\}$ to be bounded from above by a sufficiently large real number.

Proposition 3.5 *Suppose $\{X(t); t \geq 0\}$ is irreducible, and satisfies $\sup_{x \in S} q(x) < \infty$. If $\nu_1, \nu_2 \in \mathcal{V}_E^c$ satisfy $\nu_1 = \nu_2$ on M_0 , then $\nu_1 = \nu_2$ on $M_0 \cup E$.*

Proof Recall that if $\{X(t); t \geq 0\}$ is irreducible, so is $\{X_n\}_{n \geq 0}$. Given $\nu_1, \nu_2 \in \mathcal{V}_E^c$, define the vectors μ_1 and μ_2 , where for each $i \in \{1, 2\}$,

$$\mu_i(y) := q(y)\nu_i(y), \quad y \in M_0 \cup E.$$

It is easy to see μ_1 and μ_2 are elements of $\mathcal{G}_E$, as we are assuming $\sup_{y \in S} q(y) < \infty$. Finally, since $\nu_1(z) = \nu_2(z)$ for each $z \in M_0$, it follows that $\mu_1(z) = \mu_2(z)$ for each $z \in M_0$, which implies by Proposition 2.5 that $\mu_1 = \mu_2$, and so $\nu_1 = \nu_2$, proving the claim. $\diamond$

The following corollary is an immediate consequence of these results.

Corollary 3.1 *The vectors $\{\nu_M^{(x)}\}_{x \in M_0}$ form a basis of $\mathcal{V}_E^c$.*

We close this section by mentioning that another important line of research related to the ideas in this section consists of the works of Adan et al. [2], Adan and Resing [1], and Selen et al. [19]. In all three of these papers, the authors show that each ergodic CTMC within a class of Markov processes (where the class of Markov processes studied in [1] subsumes the class studied in [2], and the class studied in [19] subsumes the class studied in [1]) admits a stationary distribution that can be expressed as a finite sum of geometric product-form terms over all but finitely many states. The authors arrive at these results by first arguing that if there exists a sufficient number of linearly independent, absolutely convergent solutions to a subset of global balance equations, then over an infinitely large portion of the state space (which corresponds directly to the subset of global balance equations considered) the stationary distribution of the chain can be expressed as a linear combination of these linearly independent and absolutely convergent solutions. Once these results have been established, the authors then use the transition structures of the CTMCs

they analyze to indeed verify enough linearly independent, absolutely convergent solutions exist, and they again use the specific transition structure to argue that such solutions admit a geometric product-form structure. Contrary to [2, 1, 19], our work establishes (for irreducible, positive recurrent Markov chains) the existence of a sufficiently large enough number of linearly independent, absolutely convergent solutions, *independently of the transition structure of the chain!* Having said this, however, it appears to be impossible to say something more concrete about the *structure* of such solutions (i.e. being able to conclude the solutions all admit a geometric product-form structure, which obviously will not be true for any chain) without knowing more about the transition structure of the CTMC under study. Even though, unlike [2, 1, 19], our results do not explicitly identify new classes of ergodic CTMCs having stationary distributions that exhibit sufficiently tractable representations, what our results do suggest is that the program used in [2, 1, 19] can potentially be extended in many very interesting ways to study the stationary distribution of a much broader class of CTMCs.

4 Applications and Connections to Previous Ideas/Results

Our main objective in this section is to illustrate the importance of the theory developed in Sections 2 and 3. The main takeaway from this paper is that it is arguably much more useful to focus on calculating an invariant measure of a Markov chain instead of its stationary distribution, as focusing on the invariant measure makes the task of establishing iterative methods, as well as convergence of those iterative methods, much easier for a larger class of chains. Moreover, once an invariant measure has been found, the theory dictates that the stationary distribution is simply a normalized version of that invariant measure.

4.1 Iterative Methods and Google PageRank

We begin this section by connecting the ideas briefly discussed in the Introduction to our results from Section 2. Suppose $\{X_n\}_{n \geq 0}$ is an irreducible, finite-state DTMC, and fix a state $x \in S$. Setting $E := \{x\}^c$ and $M := E^c = \{x\}$, it is clear that $\mathcal{G}_{E;x}$ consists of the set of all

equations of the form

$$\gamma(y) = p(x, y) + \sum_{z \in E} \gamma(z) p(z, y), \quad y \in E$$

and due to $\{X_n\}_{n \geq 0}$ being positive recurrent, $\mu_M^{(x)} = \mu^{(x)}$ is the unique solution of $\mathcal{G}_{E;x}$. Not only that, $\mu^{(x)}$ appears as the pointwise limit of the sequence of vectors $\{\gamma_m^{(x)}\}_{m \geq 0}$, which satisfy the following recursive scheme. Expressing each $\gamma_m^{(x)}$ vector in block-partitioned form, it is clear that $\gamma_0^{(x)}[\{x\}^c] = \mathbf{0}$, and for each integer $m \geq 0$,

$$\gamma_{m+1}^{(x)}[\{x\}^c] = \mathbf{P}[\{x\}, \{x\}^c] + \gamma_m^{(x)}[\{x\}^c] \mathbf{P}[\{x\}^c, \{x\}^c]$$

which provides us with a natural iterative method for calculating $\mu^{(x)}[\{x\}^c]$, and therefore $\mu^{(x)}$, which in turn can be normalized to get π .

It is worth reemphasizing that the above iterative scheme (i) is valid even when $\{X_n\}_{n \geq 0}$ is not aperiodic, (ii) all calculations required with this method only involve addition and multiplication of nonnegative real numbers, and (iii) the vectors $\gamma_{m, \{x\}^c}^{(x)}$ are pointwise non-decreasing in m : the classical power iteration method possesses none of these features. Furthermore, aside from this iterative procedure, the real goal is to find the unique solution $\mu^{(x)}[\{x\}^c]$ of

$$\mu^{(x)}[\{x\}^c] = \mathbf{P}[\{x\}, \{x\}^c] + \mu^{(x)}[\{x\}^c] \mathbf{P}[\{x\}^c, \{x\}^c]$$

and given $\mathbf{P}[\{x\}^c, \{x\}^c]$ is a substochastic matrix when $\{X_n\}_{n \geq 0}$ is irreducible, there are many existing numerical procedures, such as Krylov subspace methods and their variants (see e.g. Trefethen and Bau [20]) that can be used to solve this linear system. Note that finding the stationary distribution of the random surfer DTMC behind the Google PageRank algorithm involves finding the unique solution to a linear system involving a substochastic transition probability matrix, but here we have shown that, in order to calculate the stationary distribution of *any* irreducible, finite-state DTMC, it still suffices to solve a linear system with a substochastic matrix.

4.2 Decomposition Results for DTMCs

Suppose $\{X_n\}_{n \geq 0}$ is an irreducible, positive recurrent DTMC, having a countable state space S and a transition probability matrix $\mathbf{P}$. We assume S is of the form

$$S = M \cup \left[\bigcup_{k=1}^n E_k \right]$$

where the sets $M, E_1, E_2, \dots, E_n$ form a partition of S . In light of this partition, we assume $\mathbf{P}$ satisfies the following property: for each $j, k \in [n] := \{1, 2, \dots, n\}$ where $j \neq k$, each $x \in E_j$, and each $y \in E_k$, $p(x, y) = 0$. In other words, it is impossible for $\{X_n\}_{n \geq 0}$ to make a one-step transition from a state in E_j to a state in E_k , so we refer to M as the *middle* of the state space, and refer to $E_1, E_2, \dots, E_n$ as the *edges* of the state space. Furthermore, for each $k \in [n]$, we define

$$M_k := \{x \in M : \sup_{y \in E_k} p(x, y) > 0\}$$

as the set of all states that $\{X_n\}_{n \geq 0}$ can reach E_k in one step.

The following result provides us with a simple, yet interesting representation for the invariant measure of $\{X_n\}_{n \geq 0}$ generated by a state $x \in M$.

Theorem 4.1 *Suppose $x \in M$. Then the invariant measure $\mu^{(x)}$ is as follows.*

(i) *For each $k \in [n]$, and each $y \in E_k$,*

$$\mu^{(x)}(y) = \sum_{z \in M_k} \mu^{(x)}(z) \mu_M^{(z)}(y).$$

(ii) *For each $w \in M$, where $w \neq x$,*

$$\mu^{(x)}(w) = \sum_{z \in M} \mu^{(x)}(z) p(z, w) + \sum_{k=1}^n \sum_{z \in M_k} \sum_{y \in E_k} \mu^{(x)}(z) \mu_M^{(z)}(y) p(y, w).$$

(iii) *Finally, the row vector $[\mu^{(x)}(z)]_{z \in M}$ is the minimal nonnegative solution among the set of all*

vectors $[\gamma^{(x)}(z)]_{z \in M}$ satisfying $\gamma^{(x)}(x) = 1$, and for each $w \neq x$,

$$\gamma^{(x)}(w) = \sum_{z \in M} \gamma^{(x)}(z)p(z, w) + \sum_{k=1}^n \sum_{z \in M_k} \sum_{y \in E_k} \gamma^{(x)}(z)\mu_M^{(z)}(y)p(y, w).$$

The $[\mu^{(x)}(z)]_{z \in M}$ terms also appear as the limit of a sequence of vectors $[\gamma_m^{(x)}(w)]_{w \in M}$, $m \geq 0$, where $\gamma_0^{(x)}(x) = 1$, $\gamma_0^{(x)}(w) = 0$ for each $w \in M \setminus \{x\}$, and for each integer $m \geq 0$, $\gamma_{m+1}^{(x)}(x) = 1$, and

$$\gamma_{m+1}^{(x)}(w) = \sum_{z \in M} \gamma_m^{(x)}(z)p(z, w) + \sum_{k=1}^n \sum_{z \in M_k} \sum_{y \in E_k} \gamma_m^{(x)}(z)\mu_M^{(z)}(y)p(y, w)$$

for each $w \in M \setminus \{x\}$.

Proof Fix $k \in [n]$: for each state $y \in E_k$, we have

$$\begin{aligned} \mu^{(x)}(y) &= \mathbb{E}_y \left[\mathbf{1}_{\{\tilde{\sigma}_x < \infty\}} \prod_{\ell=1}^{\tilde{\sigma}_x} \frac{p(\tilde{X}_\ell, \tilde{X}_{\ell-1})}{\tilde{p}(\tilde{X}_{\ell-1}, \tilde{X}_\ell)} \right] \\ &= \sum_{z \in M} \mathbb{E}_y \left[\mathbf{1}_{\{\tilde{\sigma}_x < \infty, \tilde{X}_{\tilde{\sigma}_x} = z\}} \prod_{\ell=1}^{\tilde{\sigma}_x} \frac{p(\tilde{X}_\ell, \tilde{X}_{\ell-1})}{\tilde{p}(\tilde{X}_{\ell-1}, \tilde{X}_\ell)} \right] \\ &= \sum_{z \in M} \mu^{(x)}(z)\mu_M^{(z)}(y) \end{aligned}$$

proving (i).

Next, fix a state $w \in M \setminus \{x\}$: conditioning on $\tilde{X}_1$ gives

$$\begin{aligned} \mu^{(x)}(w) &= \sum_{z \in M} \mu^{(x)}(z)p(z, w) + \sum_{k=1}^n \sum_{y \in E_k} \mu^{(x)}(y)p(y, w) \\ &= \sum_{z \in M} \mu^{(x)}(z)p(z, w) + \sum_{k=1}^n \sum_{y \in E_k} \sum_{z \in M} \mu^{(x)}(z)\mu_M^{(z)}(y)p(y, w) \end{aligned}$$

proving (ii).

It remains to establish (iii). First, define $\tilde{\Omega}_M^{(x)}$ as the number of times $\{\tilde{X}_n\}_{n \geq 0}$ makes a transition to a state in M , up to (and including) the instant at which it finally reaches state

$x \in M$. Next, we define, for each $w \in M$, and each integer $m \geq 0$,

$$\mu_{M;m}^{(x)}(w) := \mathbb{E}_y \left[\mathbf{1}_{\{\tilde{\sigma}_x < \infty, \tilde{\Omega}_M^{(x)} \leq m\}} \prod_{\ell=1}^{\tilde{\sigma}_x} \frac{p(\tilde{X}_\ell, \tilde{X}_{\ell-1})}{\tilde{p}(\tilde{X}_{\ell-1}, \tilde{X}_\ell)} \right].$$

Obviously $\mu_{M;m}^{(x)}(x) = 1$ for each $m \geq 0$, and $\mu_{M;0}^{(x)}(w) = 0$ for each $w \in M \setminus \{x\}$. Moreover, for each integer $m \geq 0$, and each $w \in M \setminus \{x\}$,

$$\mu_{M;m+1}^{(x)}(w) = \sum_{z \in M} \mu_{M;m}^{(x)}(z) p(z, w) + \sum_{k=1}^n \sum_{y \in E_k} \sum_{z \in M} \mu_{M;m}^{(x)}(z) \mu_M^{(z)}(y) p(y, w)$$

and from this observation, it remains to show $[\delta_m^{(x)}(w)]_{w \in M} = [\mu_{M;m}^{(x)}(w)]_{w \in M}$, and doing so is straightforward. $\diamond$

Feinberg and Chiu [6] present a method for calculating the stationary distribution of a finite-state DTMC exhibiting the same type of structure, except there each set M_k consists of a single state, meaning there is only way for the chain to enter edge set E_k , for each $k \in [n]$. Feinberg and Chiu also argue in [6] that the state space of any irreducible, finite-state DTMC can be enlarged in order to make the resulting DTMC have this type of structure, but the transition probabilities of the resulting DTMC contain elements from the stationary distribution, which means another type of iterative argument has to be used in order to calculate the stationary distribution. This scheme is rather complicated, and indeed, the authors of [6] were not successful in rigorously establishing convergence for any irreducible, finite-state chain.

It may also be useful to apply Theorem 4.1 when $n = 1$, i.e. when there is only one edge set, which corresponds to the decomposition considered in Section 2. When this is done, $|M_0|$ measures of the form $\mu_M^{(x)}$, $x \in M_0$, have to be computed, but all of these measures satisfy a linear system having as its square matrix $\mathbf{P}[E_1, E_1]$. There is a natural tradeoff to consider here: while a relatively large M will require us to calculate more $\mu_M^{(x)}$ vectors, at the same time it may actually be easier to find these quantities due to E_1 getting smaller in size, although we highly suspect that E_1 should contain many more elements than M in any application of this result.

4.3 Decomposition Results for CTMCs

This project was initially motivated by the work of Mélykúti and Pfaffelhuber [16], where the authors study the following problem: given two irreducible, finite-state continuous-time Markov chains, if we ‘glue’ the two chains together at two states in a well-defined manner (this notion of gluing will be discussed in Section 4) to what extent can the elements of the stationary distribution of the glued chain be expressed in terms of the elements of the stationary distributions of the two simpler chains? The aforementioned paper appears to have been motivated by previous work of Mélykúti [15], which addresses the problem of gluing together two finite-state continuous-time Markov chains at one state: in [15, 16], the authors are interested in applying their ideas toward the study of stochastic biochemical reaction networks. Moreover, in the work of Meng et al. [17], the authors discuss how to leverage the results found in [15, 16] to study the stationary distributions of more elaborate chemical reaction networks.

In this subsection, we address a generalization of a question posed in [16]. Fix an integer $n \geq 1$, and consider a collection of pairwise disjoint sets $M, E_1, E_2, \dots, E_n$, where each set in the collection is assumed to be finite. Associated with each integer $k \in \{1, 2, \dots, n\}$ is an irreducible CTMC $\{X^{(k)}(t); t \geq 0\}$, having state space $S_k := M \cup E_k$ and transition matrix $\mathbf{Q}_k := [q_k(x, y)]_{x, y \in S_k}$. These CTMCs are all positive recurrent, and for each integer $k \in \{1, 2, \dots, n\}$, we let $\mathbf{p}_k$ represent the stationary distribution of $\{X^{(k)}(t); t \geq 0\}$. Letting $\{X_n^{(k)}\}_{n \geq 0}$ denote the embedded DTMC associated with $\{X^{(k)}(t); t \geq 0\}$, we can define the measures $\mu_{M,k}^{(x)} := [\mu_{M,k}^{(x)}(y)]_{y \in M \cup E_k}$ and $\nu_{M,k}^{(x)} := [\nu_{M,k}^{(x)}(y)]_{y \in M \cup E_k}$, for $x \in M$, where for each $y \in M \cup E_k$,

$$\mu_{M,k}^{(x)}(y) := \mathbb{E}_x \left[\sum_{n=0}^{\eta_M^{(k)}-1} \mathbf{1}_{\{X_n^{(k)}=y\}} \right], \quad \nu_{M,k}^{(x)}(y) := q_k(x) \mathbb{E}_x \left[\int_0^{\tau_M^{(k)}} \mathbf{1}_{\{X^{(k)}(t)=y\}} dt \right].$$

These n CTMCs can be ‘glued together’ to form a new CTMC $\{X(t); t \geq 0\}$ which we will refer to throughout as the ‘glued chain’, whose state space is given by $S := M \cup E$, where $E := \cup_{k=1}^n E_k$. The transition rate matrix $\mathbf{Q} = [q(x, y)]_{x, y \in S}$ of $\{X(t); t \geq 0\}$ is

defined as follows:

$$q(x, y) = \sum_{k=1}^n q_k(x, y) \mathbf{1}_{\{x \in S_k, y \in S_k\}}, \quad x, y \in S.$$

It is not difficult to see that $\{X(t); t \geq 0\}$ is irreducible, and since S is a finite set, $\{X(t); t \geq 0\}$ has a unique stationary distribution $\mathbf{p}$. Likewise, we can associate with the glued chain and its embedded DTMC $\{X_n\}_{n \geq 0}$ the measures $\mu_M^{(x)} := [\mu_M^{(x)}(y)]_{y \in M \cup E}$ and $\nu_M^{(x)} := [\nu_M^{(x)}(y)]_{y \in M \cup E}$, for $x \in M$, where for each $y \in M \cup E$,

$$\mu_M^{(x)}(y) := \mathbb{E}_x \left[\sum_{n=0}^{\eta_M-1} \mathbf{1}_{\{X_n=y\}} \right], \quad \nu_M^{(x)}(y) := q(x) \mathbb{E}_x \left[\int_0^{\tau_M} \mathbf{1}_{\{X(t)=y\}} dt \right].$$

The main question we address in this section is as follows: to what extent can the elements of $\mathbf{p}$ be expressed in terms of the elements of the stationary distributions $\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_n$ associated with the chains used to create the glued chain? Mélykúti and Pfaffelhuber [16] answer this question for the case where both $n = 2$ and $|M| = 2$, but we will soon find that their results can be extended in a very natural way for the case where n is an arbitrary positive integer and M is an arbitrary finite set, by leveraging many of the ideas from the previous sections, as well as some ideas from a subarea of applied probability often referred to as the theory of matrix-analytic methods.

Proposition 4.1 *For each $y \in E_k$,*

$$p(y) = \sum_{z \in M_k} p(z) \nu_{M,k}^{(z)}(y) \tag{10}$$

Moreover,

$$p_k(y) = \sum_{z \in M_k} p_k(z) \nu_{M,k}^{(z)}(y). \tag{11}$$

Readers should keep in mind that Proposition 4.1 is still valid when some/all of the sets $M, E_1, \dots, E_n$ are countably infinite.

Proof Using a result from Section 5.5 of Latouche and Ramaswami [13], we find that for

each $y \in E_k$,

$$p(y) = \sum_{x \in M} p(x)q(x)\mathbb{E}_x \left[\int_0^{\tau_M} \mathbf{1}_{\{X(t)=y\}} dt \right] = \sum_{x \in M_k} p(x)q(x)\mathbb{E}_x \left[\int_0^{\tau_M} \mathbf{1}_{\{X(t)=y\}} dt \right] \quad (12)$$

where the second equality follows from

$$\mathbb{E}_x \left[\int_0^{\tau_M} \mathbf{1}_{\{X(t)=y\}} dt \right] = 0$$

for each $x \in M \setminus M_k$, as it is impossible for $\{X(t); t \geq 0\}$ to make a one-step transition from x to any state in E_k . Likewise, the same line of reasoning also yields

$$p_k(y) = \sum_{x \in M_k} p_k(x)q_k(x)\mathbb{E}_x \left[\int_0^{\tau_M^{(k)}} \mathbf{1}_{\{X_k(t)=y\}} dt \right] \quad (13)$$

where $\tau_M^{(k)} := \inf\{t \geq 0 : X^{(k)}(t-) \neq X^{(k)}(t) \in M\}$. Readers should also note that both (10) and (11) also follow from Theorem 4.1(i), combined with Proposition 3.1 and 3.2. Indeed, to prove (10), observe that for each $y \in E_k$, and each $x \in M_k$,

$$\nu_M^{(x)}(y) = \frac{q(x)}{q(y)}\mu_M^{(x)}(y) = \sum_{z \in M_k} \frac{q(x)}{q(z)}\mu^{(x)}(z)\frac{q(z)}{q(y)}\mu_M^{(z)}(y) = \sum_{z \in M_k} \nu^{(x)}(z)\nu_M^{(z)}(y). \quad (14)$$

A similar line of reasoning can be used to establish (11).

Finally, observe that for each $x \in M_k$, conditioning on the first jump of $\{X^{(k)}(t); t \geq 0\}$ gives

$$\begin{aligned} q_k(x)\mathbb{E}_x \left[\int_0^{\tau_M^{(k)}} \mathbf{1}_{\{X^{(k)}(t)=y\}} dt \right] &= \sum_{z \in E_k} q_k(x, z)\mathbb{E}_z \left[\int_0^{\tau_M^{(k)}} \mathbf{1}_{\{X^{(k)}(t)=y\}} dt \right] \\ &= \sum_{z \in E_k} q(x, z)\mathbb{E}_z \left[\int_0^{\tau_M} \mathbf{1}_{\{X(t)=y\}} dt \right] \\ &= q(x)\mathbb{E}_x \left[\int_0^{\tau_M} \mathbf{1}_{\{X(t)=y\}} dt \right] \end{aligned}$$

and after applying this fact to (14), we arrive at (10). $\diamond$

Even though Proposition 4.1 follows relatively quickly from a known result, this result

is arguably not at all well-known outside of the matrix-analytic methods literature, and the main goals of the authors of [16] are rather separate from the main goal of most, if not all papers featuring usage of ideas from the matrix-analytic methods literature. What makes both Theorem 1 from [16] and Proposition 4.1 novel is that, for each $k \in \{1, 2, \dots, n\}$ and each $y \in E_k$, both $p(y)$ and $p_k(y)$ can be expressed in terms of the same quantities $\{\nu_{M,k}^{(z)}(y)\}_{z \in M_k, y \in E_k}$, and in turn, each $\nu_{M,k}^{(z)}$ is the unique solution of a linear system: uniqueness is guaranteed due to each E_k being a finite set. This observation, we feel, best captures what the authors of [16] were looking for when they came up with their main result.

We close the paper by showing how the $p(y)$ terms, for each $y \in M$, can be expressed in a manner analogous to how these probabilities (for the case where $|M| = 2$ and $n = 2$) are expressed in [16], although a simple modification of the idea behind Theorem 4.1 can also be used to find $\mathbf{p}$. The overall scheme we use to find these terms is essentially the same as that given in subsection 3.3.3 on page 45 of [9].

Fix $x_0 \in M$, and recall that the row vector $\nu^{(x_0)} := \{\nu^{(x_0)}(y)\}_{y \in S}$, defined as

$$\nu^{(x_0)}(y) := q(x_0) \mathbb{E}_{x_0} \left[\int_0^{\tau_{x_0}} \mathbf{1}_{\{X(t)=y\}} dt \right], \quad y \in S$$

is a positive scalar multiple of the stationary distribution $\mathbf{p}$ of $\{X(t); t \geq 0\}$. More particularly, $\mathbf{p} = p(x_0) \nu^{(x_0)}$. The key to calculating the $\nu^{(x_0)}(y)$ terms, for $y \in M$, involves gaining an understanding of the following matrices. Let $E := \bigcup_{k=1}^n E_k$, and define the matrices $\mathbf{X} = [x_{z,w}]_{z,w \in M \setminus \{x_0\}}$ and $\mathbf{Y} = [y_{z,w}]_{z,w \in M \setminus \{x_0\}}$, where for each $z, w \in M \setminus \{x_0\}$,

$$x_{z,w} := \mathbb{E}_z \left[\int_0^{\tau_{E \cup \{x_0\}}} \mathbf{1}_{\{X(t)=w\}} dt \right], \quad y_{z,w} := \mathbb{E}_z \left[\int_0^{\tau_{x_0}} \mathbf{1}_{\{X(t)=w\}} dt \right].$$

and let $\mathbf{Q}_{M \setminus \{x_0\}} := [q(z, w)]_{z,w \in M \setminus \{x_0\}}$ denote the submatrix of $\mathbf{Q}$ where the rows and columns of $\mathbf{Q}_{M \setminus \{x_0\}}$ correspond to the elements from $M \setminus \{x_0\}$. The next lemma shows that $\mathbf{X}$ can be expressed in terms of $\mathbf{Q}_{M \setminus \{x_0\}}$, an invertible matrix. Throughout, we will let $\mathbf{I}$ denote an identity matrix, where the number of rows and columns of $\mathbf{I}$ will be clear from the surrounding context.

Lemma 4.1 $\mathbf{Q}_{M \setminus \{x_0\}}$ is an invertible matrix, and $\mathbf{X} = (-\mathbf{Q}_{M \setminus \{x_0\}})^{-1}$.

Proof This result is well-known, so we omit the proof. $\diamond$

The next lemma shows how to express $\mathbf{Y}$ in terms of $\mathbf{X}$. In order to do so, we introduce the matrices $\{\mathbf{A}_k\}_{k=1}^n$, where for each $k \in \{1, 2, \dots, n\}$, the matrix $\mathbf{A}_k := [a_{z,w}^{(k)}]_{z,w \in M \setminus \{x_0\}}$ consists of the elements

$$a_{z,w}^{(k)} := \sum_{r \in E_k} \mathbb{P}_z(X(\tau_{E \cup \{x_0\}}) = r) \mathbb{P}_r(X(\tau_M) = w), \quad z, w \in M \setminus \{x_0\}.$$

A close look at $a_{z,w}^{(k)}$ reveals it represents that probability that, given $\{X(t); t \geq 0\}$ occupies state z at time zero, the chain will reach a state outside of M before reaching $\{x_0\}$, and when it again returns to M , it does so by visiting state w . Each element of $\mathbf{A}_k$ consists of quantities that can be solved for using a standard first-step analysis argument, the details of which we leave to the interested reader.

Lemma 4.2 *The matrix $\mathbf{Y}$ is invertible, and satisfies $\mathbf{Y} = (\mathbf{I} - \sum_{k=1}^n \mathbf{A}_k)^{-1} \mathbf{X}$.*

Proof Fix $z, w \in M \setminus \{x_0\}$, where possibly $z = w$. Then

$$\begin{aligned} y_{z,w} &= \mathbb{E}_z \left[\int_0^{\tau_{x_0}} \mathbf{1}_{\{X(t)=w\}} dt \right] \\ &= \mathbb{E}_z \left[\int_0^{\tau_{E \cup \{x_0\}}} \mathbf{1}_{\{X(t)=w\}} dt \right] + \mathbb{E}_z \left[\int_{\tau_{E \cup \{x_0\}}}^{\tau_{x_0}} \mathbf{1}_{\{X(t)=w\}} dt \right] \\ &= x_{z,w} + \sum_{k=1}^n \sum_{r \in E_k} \left(\mathbb{P}_z(X(\tau_{E \cup \{x_0\}}) = r) \mathbb{E}_r \left[\int_0^{\tau_{x_0}} \mathbf{1}_{\{X(t)=w\}} dt \right] \right). \end{aligned}$$

Now observe that for each $r \in E_k$, and each $k \in \{1, 2, \dots, n\}$, we have

$$\begin{aligned} \mathbb{E}_r \left[\int_0^{\tau_{x_0}} \mathbf{1}_{\{X(t)=w\}} dt \right] &= \mathbb{E}_r \left[\int_{\tau_M}^{\tau_{x_0}} \mathbf{1}_{\{X(t)=w\}} dt \right] \\ &= \sum_{v \in M \setminus \{x_0\}} \mathbb{P}_r(X(\tau_M) = v) y_{v,w} \end{aligned}$$

so that we may write

$$y_{z,w} = x_{z,w} + \sum_{k=1}^n \sum_{r \in E_k} \sum_{v \in M \setminus \{x_0\}} \mathbb{P}_z(X(\tau_{E \cup \{x_0\}}) = r) \mathbb{P}_r(X(\tau_M) = v) y_{v,w}.$$

Given this statement is true for all $z, w \in M \setminus \{x_0\}$, we can more elegantly state all of the

resulting equations together in matrix form as follows:

$$\mathbf{Y} = \mathbf{X} + \sum_{k=1}^n \mathbf{A}_k \mathbf{Y}$$

or, equivalently,

$$\left(\mathbf{I} - \sum_{k=1}^n \mathbf{A}_k \right) \mathbf{Y} = \mathbf{X}.$$

Finally, since $\mathbf{X}$ is invertible, both $(\mathbf{I} - \sum_{k=1}^n \mathbf{A}_k)$ and $\mathbf{Y}$ are as well, and solving for $\mathbf{Y}$ gives the claim. $\diamond$

The last crucial step involves calculating the row vector $\boldsymbol{\nu}_{M \setminus \{x_0\}}^{(x_0)} := [\nu^{(x_0)}(w)]_{w \in M \setminus \{x_0\}}$. This vector can be expressed in terms of $\mathbf{Y}$, as well as the row vectors

$$\mathbf{q}_{x_0, M \setminus \{x_0\}} := [q(x_0, z)]_{z \in M \setminus \{x_0\}} \quad \mathbf{b}_k = [b_z^{(k)}]_{z \in M \setminus \{x_0\}}$$

where for each $z \in M \setminus \{x_0\}$,

$$b_z^{(k)} := \sum_{r \in E_k} q(x_0, r) \mathbb{P}_r(X(\tau_M) = z).$$

Again, each element of $\mathbf{b}_k$ can be calculated using a standard first-step analysis argument.

Proposition 4.2 *The row vector $\boldsymbol{\nu}_{M \setminus \{x_0\}}^{(x_0)}$ satisfies $\boldsymbol{\nu}_{M \setminus \{x_0\}}^{(x_0)} = (\mathbf{q}_{x_0, M \setminus \{x_0\}} + \sum_{k=1}^n \mathbf{b}_k) \mathbf{Y}$.*

Proof Fix $w \in M \setminus \{x_0\}$. Then

$$\begin{aligned} \nu^{(x_0)}(w) &= \sum_{z \in M \setminus \{x_0\}} q(x_0, z) \mathbb{E}_z \left[\int_0^{\tau_{x_0}} \mathbf{1}_{\{X(t)=w\}} dt \right] + \sum_{k=1}^n \sum_{r \in E_k} q(x_0, r) \mathbb{E}_r \left[\int_0^{\tau_{x_0}} \mathbf{1}_{\{X(t)=w\}} dt \right] \\ &= \sum_{z \in M \setminus \{x_0\}} q(x_0, z) y_{z,w} + \sum_{k=1}^n \sum_{r \in E_k} \sum_{v \in M \setminus \{x_0\}} q(x_0, r) \mathbb{P}_r(X(\tau_M) = v) y_{v,w} \\ &= \sum_{z \in M \setminus \{x_0\}} q(x_0, z) y_{z,w} + \sum_{k=1}^n \sum_{v \in M \setminus \{x_0\}} b_v^{(k)} y_{v,w}. \end{aligned}$$

This implies the matrix equality

$$\nu_{M \setminus \{x_0\}}^{(x_0)} = \mathbf{q}_{x_0, M \setminus \{x_0\}} \mathbf{Y} + \sum_{k=1}^n \mathbf{b}_k \mathbf{Y} = \left(\mathbf{q}_{x_0, M \setminus \{x_0\}} + \sum_{k=1}^n \mathbf{b}_k \right) \mathbf{Y}$$

which is what we set out to prove. $\diamond$

Finally, note that the normalizing equation can be used to find $p(x_0)$:

$$p(x_0) = \left(\sum_{z \in M} \nu^{(x_0)}(z) + \sum_{k=1}^n \sum_{w \in M_k} \sum_{r \in E_k} \nu^{(x_0)}(w) \nu_{M,k}^{(w)}(r) \right)^{-1}$$

where readers should recall that $\nu^{(x_0)}(x_0) = 1$. Together, Propositions 4.1 and 4.2 cover a significant generalization of Theorem 1 from [16].

5 Open Questions

We believe that our main results raise a number of interesting questions. First, our main results show that focusing on calculating an invariant measure of a Markov chain may be advantageous relative to calculating the stationary distribution of said chain directly, as the iterative scheme we present for calculating an invariant measure satisfies a nice monotonicity property, and involves addition and multiplication of nonnegative terms. Keep in mind too that each recurrent state corresponds to an invariant measure, meaning it may be possible to set up useful algorithms that consist of calculating invariant measures associated with multiple states in parallel, which may aid in finding better stopping criteria for iterative algorithms. Another possible research question could involve studying whether or not it is worth modifying the RLGL method of [3] to calculate $\mu^{(x)}$ instead of the stationary distribution directly.

Another question involves studying when, for an arbitrary irreducible, finite-state chain, it is beneficial to let M consist of two or more states: doing so requires us to solve (at most) $|M|$ linear systems of equations, then use those solutions to solve for the terms associated with states in M of an invariant measure associated with a fixed state in M . Another related research direction may involve how Theorem 4.1 relates to the BlockRank ideas discussed

in Kamvar et al. [11], which improves the Google PageRank algorithm by taking advantage of blocks of linked websites that naturally occur within the structure of the internet.

Our main results are also loosely related to the work of Zheng et al. [21], as well as the work of Glynn and Zheng [8]. In these papers, the authors seek to calculate the stationary distribution $\mathbf{p}$ of an irreducible, positive recurrent DTMC by using numerical linear algebra to calculate the stationary distribution among states located in the ‘center’ of the state space, and by using simulation methods to perform calculations associated with states outside of this center. It may in fact be possible to combine our contributions with the COSIMLA technique developed in [21, 8] in order to establish more refined numerical methods for calculating various stationary expectations associated with the stationary distribution of an irreducible, positive recurrent DTMC, but at this point it is not clear (to us at least) if this is possible.

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